Ans. (d) : The member countries of International thermonuclear Experimental Reactor (ITER) are- China, the European Union, India, Japan, Korea, Russia and the United States are now engaged in a 35-year collaboration to build and operate the ITER experimental device and together bring fusion to the point where a demonstration fusion reactor can be designed will be the first fusion device to test the integrated technologies, materials, and physics regimes necessary for the commercial production of fusion-based electricity.

- If this experiment succeeds, the immediate advantage of India is that it can build fusion reactors for power generation. So, the correct answer is option (d).

57. Regarding 'DigiLocker', sometimes seen in the news, which of the following statements is/ are correct ?

1. It is a digital locker system offered by the Government under Digital India Programme.
2. It allows you to access your e-documents irrespective of your physical location.

Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c): Digilocker is one of the key initiatives under the Digital India Programme. This was released by the Department of Electronics and Information Technology (DeitY), Government of India.
The space can be utilized for storing personal documents like university certificates, permanent account number (PAN) cards, voter ID cards, and the URIs of the e-documents issued by various issuer departments.

58. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct ?

1. Other than USA and Russia, India is the only country to have launched a similar observatory into space.
2. Astrosat is a 2000 kg satellite placed in an orbit at 1650 km above the surface of the Earth.

Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d): ASTROSAT is India's first dedicated multi-wavelength space observatory. ASTROSAT will observe the universe in the optical, ultraviolet, low and high energy X-Ray regions of the electromagnetic spectrum.
India's observatory will be the fourth in space, after the Hubble (USA), Russia's Spektr R and Suzaku of Japan.
ASTROSAT with a lift-off mass of about 1513 kg, was launched into a 650 km orbit inclined at an angle of 6 degrees to the equator by PSLV-C 30.

59. Consider the following statements :

The Mangalyaan launched by ISRO

1 is also called the Mars Orbiter Mission
2. made India the second country to have a spacecraft orbit the Mars after USA.
3. made India the only country to be successful in making its spacecraft orbit the mars in its very first attempt.

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c) : The Mars orbiter Mission, also called Mangalyaan, is a space probe orbiting Mars since 24 September 2014. It was launched on 5 November, 2013 by the Indian Space Research Organization. India is the first Asian country and ISRO is the fourth space agency to send a satellite to Mars.
It is the first interplanetary mission realized by India and the first Indian spacecraft to incorporate full -scale onboard autonomy to overcome the long distances and the communication gaps due to non-visibility periods.
It is also the first Mars mission in the world to succeed mars orbit insertion in the first attempt.

60. In the context of the developments in bioinformatics, the term 'transcriptome', sometimes seen in the news, refere to-

(a) a range of enzymes used in genome editing
(b) the full range of m RNA molecules expressed by an organism
(c) the description of the mechanism of gene expression
(d) a mechanism of genetic mutation taking place in cells

Ans. (b): Bioinformatics refers to the application of computer technology to the management of biological information.
Transcriptome refers to the full range of messenger RNA (mRNA) molecules expressed from the genes of an organism.

61. 'Project Loon', sometimes seen in the news, is related to

(a) Waste management technology
(b) Wireless communication technology
(c) Solar power production technology
(d) Water conservation technology

Ans. (b): Project loon is a network of stratospheric balloons that help to connect rural and remote regions of the world to internet connectivity via wireless communication technology.

- Project loon is a research and development project being developed by Google X.
- It helps to improve communication during natural disasters to affected regions.

62. With reference to 'Bitcoins' sometimes seen in the news, which of the following statements is/are correct ?

1. Bitcoins are tracked by the Central Banks of the countries.
2. Anyone with a Bitcoin address can send and receive Bitcoins from anyone else with a Bitcoin address .
3. Online payments can be sent without either side knowing the identity of the other.

Select the correct answer using the code given below :

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans. (b) : Bitcoins can be sent directly to anyone, anywhere in the world. The only requirement is that you should have the Bitcoin address of the recipient. The process doesn't require you to reveal your identity. So, (2) and (3) are correct.
The system is a peer-to peer and transactions take place between users directly, without an intermediary. So this is not tracked by the Central bank of any intermediary. So (1) is incorrect.

ENVIRONMENT AND ECOLOGY

63. Consider the following statements :

1. The international Solar Alliance was launched at the United Nations Climate Change Conferencre in 2015.
2. The Alliance includes all the member countries of the United Nations.

Which of the statements given above is /are correct:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : The COP-21 Climate Conference was held in Paris, France, from 30th November to 12th December 2015. During this conference, India and France have launched the International Solar Alliance (ISA). ISA includes an alliance of 121 countries located between Tropic of Cancer and Tropic of Capricorn. While ISA was launched on the sidelines of the Paris summit, therefore, statement (1) is right, but it includes only 121 countries between the tropic of Capricorn and the tropic of Cancer receiving sunlight for 300 days as more. Hence statement (2) is wrong.

64. With reference to the Agreement at the UNFCCC Meeting in Paris in 2015, which of the following statements is /are correct.

1. The Agreement was signed by all the member countries of the UN and it will go into effect in 2017.
2. The Agreement aims to limit the greenhouse gas emissions so that the rise in average global temperature by the end of this century does not exceed 2^oC or even 1.5^oC above pre-industrial levels.
3. Developed countries acknowledged their historical responsibility in global warming and committed to donate $ 1000 billion a year from 2020 to help developing countries to cope with climate change.

Select the correct answer using the code given below:

(a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (b): 21st Conference of the Parties (COP 21) is also known as Paris Agreements, successfully concluded in Paris under the United Nations Framework Convention on Climate Change (UNFCCC). The agreement is scheduled to go into effect from 2020.
Statement (1) is not correct.
One of the main focuses of the agreement is to hold the increase in the global average temperature to well below 2ºC above pre-industrial level and on driving efforts to limit it even further to 1.5ºC. It covers all the crucial areas identified as essential for a comprehensive and balanced agreement, including migration adaption, loss and damage finance, technology development and transfer, capacity building and transparency of action and support. A marked departure from the past is the Agreements bottom-up approach, allowing each nation to submit its own national plan for reducing greenhouse gas emissions, rather than trying to repeat a top-down approach advocated by the Kyoto protocol, giving each country an emission reduction target.
Statement (3) is not correct.

65. Consider the following statements :

1. The Sustainable Developments Goals were first proposed in 1972 by a global think tank called the 'Club of Rome'.
2. The Sustainable Development Goals have to be achieved by 2030.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2